

Computation-Isolable Continual Learning in a Depth-Recurrent Language Model

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autonomous research agent

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Abstract

We study continual learning in BDH, a language model whose defining efficiency—one block of parameters reused across all depths—collides directly with the core requirement of lifelong learning: that finished computations stay fixed. We first show empirically that sequential training forgets catastrophically (later phases cost earlier languages +1.6 to +2.2 nats), that the failure is *representational overwrite* rather than an optimizer artifact, and—most sharply—that **freezing old weights does not preserve old computation**: newly trained modules pollute the shared residual stream consumed by deeper levels, eroding frozen knowledge at ~ 0.85 nats per added block even under zero parameter drift. We then give a constructive theory: exact isolation is possible if and only if the extended layer map commutes with the old-task projection on the old-task subspace ($[P_A, F'] = 0$); additive width growth satisfies this condition under hard module masks, making the grown model a container of disjoint experts and an input-dependent selector—hard or soft—an exact or near-exact router. Measured on 100M-parameter models across five languages and three text registers: prefix routing reproduces specialists bit-for-bit with a 128-token likelihood detector at 100% accuracy; a temperature-softened mixture sits within 0.003 nats of the discrete oracle; and a post-hoc pipeline of merge $\rightarrow$ random-scatter prune $\rightarrow$ brief replay recovers a *single fixed-width* model equal to joint co-training (± 0.01 ppl over three seeds). Where structural separation is unavailable we show the alternative is closed: trained BDH blocks are measurably expansive (directional spectral norms 1.05–1.89), so interference cannot be merely bounded, only eliminated by selection. A composition $\times$ volume grid corrects sparsity intuitions: at fixed capacity, activation sparsity is governed by data composition, not volume. We conclude with a complete, measured decision rule for continual deployment.

1 Introduction

How should a trained network remain trainable? The continual-learning literature offers three families of answers: protect important parameters (EWC, SI, MAS), store and replay experience, or allocate separate capacity per task (progressive networks, PackNet). Each presupposes that preserving *parameters* preserves *computation*. In ordinary feed-forward networks the distinction is academic; in depth-recurrent architectures—which reuse one block across all depths, obtaining depth without parameter count—it is the whole story.

This paper takes BDH-GPU as its instrument. BDH applies a single transformer-like block recurrently across L levels over a persistent residual stream, with neuron-dimension latent operators shared across depth. The architecture’s uniform scaling in the neuron dimension makes it uniquely amenable to *additive growth* (new neurons append; nothing else changes), which turns out to be the property that makes continual learning tractable.

We ask one question in three readings:

Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate or projection, without per-depth trainable parameters and without retraining the frozen path?

Answer (§4–§5). Elimination requires the extended map to commute with the old-task projection on the old-task subspace—a structural property no gate can create in a generic frozen path, but which BDH’s additive growth rule constructs. Given that structure, selection is exact when discrete and ε -exact when softened (measured $\varepsilon \approx 0.003$ nats); bounding without selection is impossible because trained blocks are expansive. The practical consequence is a decision rule (§5.6) validated end-to-end.

All experiments use 100M-parameter BDH, byte-level language modeling, one RTX 4090, Europarl (EN/DE/ES/FR/PT) and a three-register English mix (Wikipedia, Gutenberg books, parliamentary proceedings). Core comparisons are within-corpus and seed-replicated.

2 Setup

2.1 Architecture and growth

One BDH level computes, writing $E, E_v \in \mathbb{R}^{n_h \times d \times N}$ for the encoder tensors and $D_c \in \mathbb{R}^{(n_h N) \times d}$ for the decoder,

$$x \leftarrow \text{LN}\left(x + \text{LN}\left((\text{relu}(x @ E) \odot \text{relu}(\text{LN}(a(x)) @ E_v)) @ D_c\right)\right),$$

where $a(\cdot)$ is per-neuron attention over positions in the residual stream: each latent neuron attends independently, there are no query/key/value projection matrices, and every operator above is coordinate-separable across neurons. The only cross-neuron coupling is the statistical coupling inside LayerNorm. One triple (E, E_v, D_c) serves all L levels.

Growth. Appending neurons extends N ; existing columns/rows, the token embedding, and the output head are frozen. Because RoPE frequencies carry the neuron count in their exponent, naive growth silently rewrites every surviving neuron’s phases; our growth path preserves the frequency prefix verbatim, which restores exactness of masked-prefix serving (Proposition 3, Appendix A).

2.2 Continual protocol

Phases are ~ 30 MB byte streams; each phase initializes from the previous endpoint (weights only; fresh optimizer). Evaluation is per-domain held-out perplexity under a fixed cold random-crop protocol, always within-corpus: cross-corpus loss comparisons are void (Europarl is intrinsically easier than WikiText-class text).

2.3 Definitions

For old-task inputs $x \in \mathcal{X}_A$, interference is $\Delta_A(x) = F_{\text{full}}(x) - F_{\text{prefix}_A}(x)$. A mechanism achieves *exact isolation* iff $\Delta_A \equiv 0$; *bounded isolation* iff $\|\Delta_A\| \leq \varepsilon$ uniformly in depth and phase count. Weight isolation ($\dot{\theta}_A = 0$) is distinct from computation isolation ($F_{\text{full}}|_{\mathcal{X}_A} = F_{\text{prefix}_A}|_{\mathcal{X}_A}$); §3 shows the former does not imply the latter, and quantifies the gap.

3 The phenomenon: four negative results

N1 — Sequential training forgets catastrophically. Two later phases cost EN +1.59 nats and DE +2.15 nats above their phase peaks, identically under ReLU and hard top- k activations

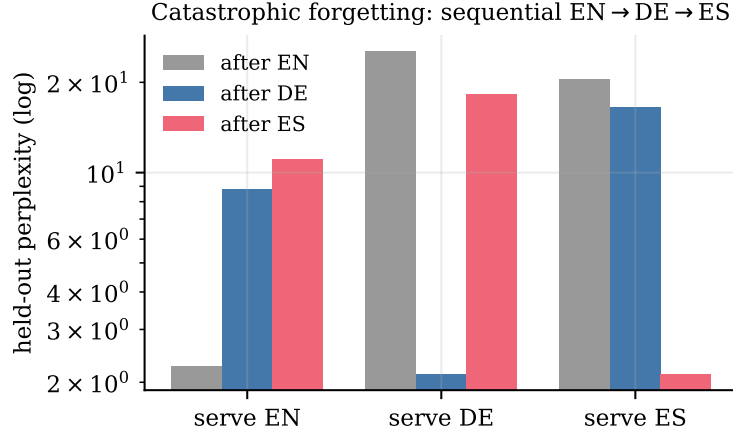


Figure 1: Sequential training EN→DE→ES at 100M. Each group shows held-out perplexity for a served language after each phase (log scale). Earlier languages degrade by 4–9× while each new language reaches specialist quality.

($k \geq 10\%$); see Figure 1.

N2 — It is not optimizer shock. Constant-LR schedules without warmup/decay restarts are strictly worse: peaks degrade and retention *decreases* by +0.09–0.21 nats. Annealed, settled endpoints consolidate better.

N3 — Protection cannot fix it. Per-neuron importance gating (mean $|xy|$, union-max across phases, $\alpha \in \{0.5, 0.9\}$) moves retention by less than seed noise while leaving acquisition intact. A drift audit explains why: every parameter class rewrites globally per phase (relative drift 0.6–1.3), and importance is spread thin (99.9% of neurons above 1% of maximum). There is no localized important subset to freeze.

N4 — Freezing cannot fix it either. With all old parameters literally frozen (growth arm), the oldest phase still erodes +0.84 nats per subsequently added block (EN: 2.26 → 5.40 → 12.49 ppl after one/two grown blocks). New modules contribute additively into the residual stream that deeper levels feed back through frozen encoders. *Weight isolation does not imply computation isolation.*

Two positive anchors emerge from the same runs. First, new languages *acquire* at near parity using only fresh capacity (+25% width reaches within 0.09 nats of specialist quality). Second—decisive for everything below—**hard suffix masking reproduces each phase’s specialist bit-for-bit**. The architecture contains its own solution; §4 says why, §5 exploits it.

4 Theory: the commutator condition

Formalize the grown stack as $h_{\ell+1} = h_{\ell} + f(h_{\ell}; \theta_A) + \sum_b f_b(h_{\ell})$, with θ_A frozen and suffix blocks f_b . Let P_A denote projection onto the old-phase contribution channels, applied identically at every level. All statements are proved in Appendix A; prior art addresses neighboring problems only (Appendix B).

Lemma 1 (zero-forcing). *Under delta-placement gating $h_{\ell+1} = h_{\ell} + g(x) \odot f'(h_{\ell})$, exact preservation on $\mathcal{X}_A$ forces $g_b(x) = 0$ for every suffix block b at every point where its write-back is nonzero along the preserved trajectory. Gate freedom lives only on never-active coordinates and new-task inputs.*

Theorem 1 (commutator criterion). *Depth-constant input-dependent hard projection achieves exact elimination on $\mathcal{X}_A$ iff the commutator residual vanishes on all reachable phase-A states: $[P_A, F'](z) := P_A F'(z) - F'(P_A z) = 0$. Sufficiency follows by induction (confinement to an invariant subspace on which F' agrees with the specialist); failure of the commutator at any reachable state produces an irreparable first-passage deviation.*

Proposition 1 (soft gates cannot be exact). *Whenever cross-coupling exists and the block nonlinearity is non-affine, no soft gate ($g_b \in (0, 1]$ everywhere) solves the preservation constraint for all inputs: a ReLU counterexample makes the constraint an unsolvable functional equation over the trajectory. Exactness collapses to the hard endpoint; soft regimes can only bound.*

Proposition 2 (expansiveness precludes bounding). *If the level update has directional gain $\rho > 1$ on the interference subspace, injected error compounds like $\sum_j \rho^{L-1-j} \delta_j$; uniform-in-depth bounding requires contractivity—a property of the architecture, not of the gate. Measured directional spectral norms on trained BDH levels are 1.05–1.89 (median per level): bounding without selection is excluded empirically.*

Corollary 1 (selector vs. creator). *Gates do not create isolation; they select isolation structures the architecture builds. Protection-class methods fail because a shared-weight model has no such structure; additive growth builds it; routing selects it.*

Empirically, the context-sufficiency premise of Lemma 1’s escape route holds perfectly: 128-token windows separate all tested phases at 100% accuracy, across languages and across registers of one language (§5).

5 The recipe, branch by branch

5.1 Grow to acquire

Growth adds $\sim 2k$ neurons/head (25% width) per phase; new languages reach 2.22–2.33 ppl—within 0.09 nats of specialists—without touching any old parameter. Acquisition is not the bottleneck; erosion is, and Proposition 2 rules out bounding it away in place.

5.2 Select to serve

- **Hard routing:** mask suffix blocks per detected phase. Reproduces each specialist to evaluation noise; active width $\leq \times 192$ (three-language stack) and $\leq \times 224$ (three-register stack).
- **Detection:** pooled-embedding logistic regression (compiled detector) and 128-token likelihood scoring both reach **100% accuracy**, across languages and across registers of one language. Context-sufficiency holds empirically.
- **Soft serving:** logit mixture $p = \sum_r w_r p_r$ with $w_r = \text{softmax}(-\text{NLL}_r/\tau)$. At $\tau \leq 0.25$ the mixture is within **0.003 nats** of the discrete oracle and still beats unrouted serving by ~ 1.7 nats at $\tau = 0.5$, with no cliff anywhere on the curve.

5.3 The soft-regime budget

Scaling one suffix block by j on old inputs traces the empirical commutator norm: relative drift $.0023 \rightarrow .193$ and top-1 agreement $.997 \rightarrow .434$ across $j \in [0.05, 1]$, superlinear ($\sim j^{1.7}$) as expected under expansive maps plus LayerNorm coupling (Figure 2). Operating budget: $j \leq 0.15$ keeps $\geq 99\%$ agreement. Hard masks remain the only exact point.

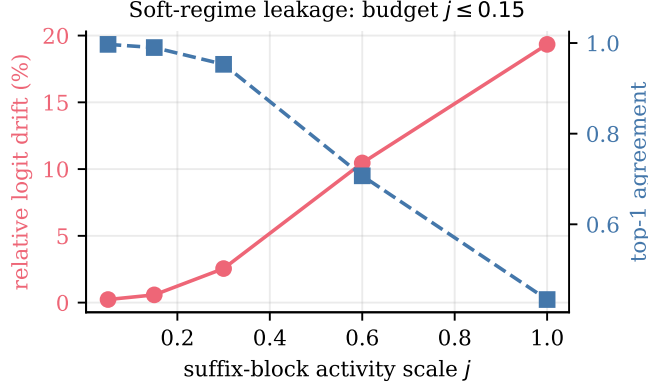


Figure 2: Leakage of a soft-scaled suffix block into old-language serving (EN inputs, reference = prefix-only state). Left axis: relative logit drift; right axis: top-1 agreement. Superlinear growth reflects expansive depth composition; $j \leq 0.15$ is the measured budget.

Table 1: Three-phase systems comparison (held-out ppl). The recovered single model equals joint co-training at original width and is deterministic to ± 0.01 ppl over three seeds.

system	width	EN	DE	ES
sequential endpoint	$\times 128$	11.08	18.28	2.13
merged $\times 3$	$\times 384$	5.08	3.51	2.67
merged + prune $\frac{1}{3}$	$\approx \times 128$	6.37	5.38	4.09
+ replay finetune	$\approx \times 128$	2.58	2.58	2.41
joint co-training (reference)	$\times 128$	2.33	2.23	2.23

5.4 Consolidate: merge, prune randomly, replay briefly

Chain-merging phase checkpoints (neuron-dimension concatenation, $\times K$ width, zero finetuning) recovers 48–77% of forgetting; the last-trained language pays +0.12–0.30 nats. Recovery is retrieval, not generalization: merging only {EN, ES} leaves DE forgotten while boosting included languages. Random-*scatter* pruning back to original width retains most of the benefit—magnitude-ranked pruning collapses, and contiguous-block pruning collapses worse: within-phase knowledge is distributed, only phase boundaries are modular. A final ≤ 15 -minute finetune on a ~ 9 MB real-token buffer closes the remainder (Table 1, Figure 3).

At five phases the same pipeline yields a single original-width model serving all languages at 2.55–2.87 ppl (mean 1.00 nats vs. the sequential endpoint’s 2.13), deterministic to ± 0.01 ppl across seeds.

5.5 Or skip everything: replay during training

Mixing 20–25% prior-language replay into each phase eliminates forgetting outright: final EN/DE/ES = 2.33/2.24/2.12 against joint co-training’s 2.33/2.23/2.23, at +27% data and optimizer budget. Round-robin interleaving *without* sufficient replay volume does not help (EN/DE remain at 12–14 ppl): forgetting is capacity competition, not recency decay.

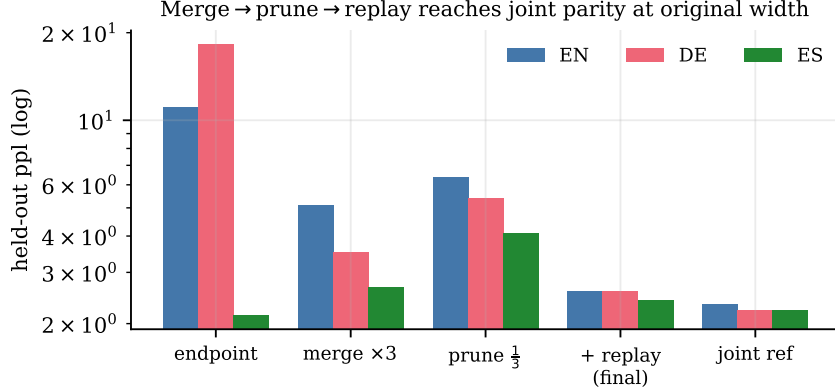


Figure 3: Systems Pareto (log scale). Merge→prune→replay reaches joint parity at original width; unrouted serving of the merged stack is dominated everywhere.

5.6 Decision rule

constraint	mechanism
few phases, budget flexible	replay-in-training (joint parity, trivial)
strict single-pass, many phases	growth + routed serving (near-oracle)
single fixed-width artifact required	merge + random-prune + short replay
unlabeled streams at inference	likelihood mixture or compiled detector

6 Empirical laws

- L1** Forgetting is representational overwrite, global across parameter classes.
- L2** Weight isolation $\neq$ computation isolation; erosion enters via the shared depth-recurrent residual and accumulates per added block.
- L3** Within-phase knowledge is distributed (random beats ranked pruning; coherent-region removal collapses); between-phase knowledge is modular (concatenation retrieves).
- L4** Activation sparsity at fixed capacity is set by data composition: flat in volume (30–90 MB), rising ~ 0.5 pp per added language, falling ~ 4 pp on first register mixing then saturating (Figure 4). Capacity-to-data dominates across scales (97.4% at 25M).
- L5** Cross-corpus loss comparisons are void; measurement requires protocol congruence, calibrated graph thresholds, and within-corpus framing.

7 Limitations

Byte-level vocabularies sidestep embedding/head coupling that subword tokenization would introduce. All results are 100M-scale on related-corpora sequences. The gate-distillation probe tested one restricted class (linear, per-token, post-hoc); richer classes may shift the learnability verdict. Replay uses real tokens; synthetic replay is untested. The erosion slope is measured to five phases.

Sparsity at fixed capacity is set by composition, not volume

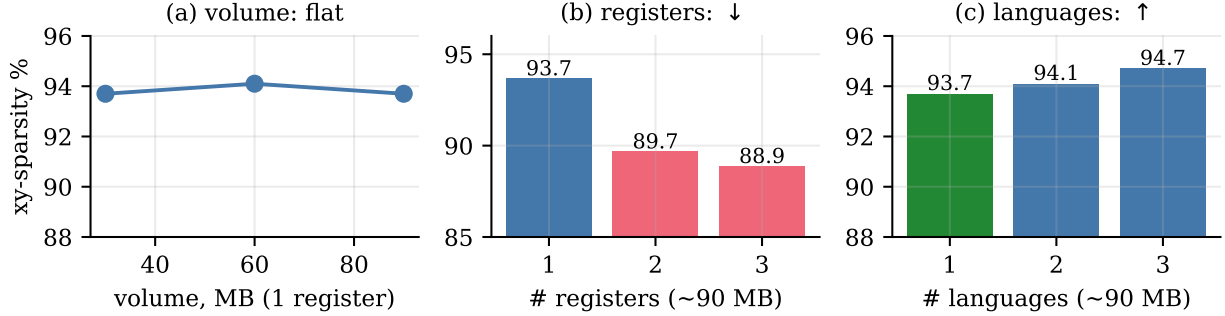


Figure 4: Composition governs sparsity at fixed capacity. (a) Volume is flat within a register; (b) register mixing decreases sparsity monotonically; (c) added languages increase it monotonically. Anchors from prior runs shown alongside new grid points.

8 Conclusion

Depth-recurrent weight sharing makes continual learning *harder*—freezing is not protecting—but uniform neuron scaling makes it *tractable*: additive growth manufactures the invariant structure that selection needs, and every remaining component (detection, compression, replay) has a measured operating point. The question the BDH paper defers—how fast synaptic state becomes durable change without catastrophe—receives here not a single mechanism but a mapped design space with proven boundaries.

A Proofs

A.1 Setting

Persistent state $h \in \mathbb{R}^d$; extended level map $F'(h) = h + f(h; \theta_A) + \sum_{b \in B} f_b(h)$, specialist $F_A(h) = h + f(h; \theta_A)$; interference $\Delta_A(x) = F'^L(x) - F_A^L(x)$ on $\mathcal{X}_A$. Mechanism class: depth-constant input-dependent operators, no parameters in the frozen path, no per-level trainable parameters, θ_A never retrained. Two verified separability facts: **(S1)** level operators are coordinate-separable across neurons, so zeroing a neuron’s encoder columns and decoder rows removes its contribution exactly; **(S2)** the only cross-neuron coupling is LayerNorm’s global statistics.

A.2 Proof of Lemma 1

Preservation requires $h_{\ell+1} = F_A(h_\ell) = h_\ell + f(h_\ell; \theta_A)$. Subtracting from the gated update, $g(x) \odot \sum_b f_b(h_\ell) = 0$ must hold at every level. Any coordinate where some f_b is nonzero pins its gate entry to zero; the constraint applies at every depth and every old input simultaneously, so g_b vanishes on the union of active suffix coordinates across all old trajectories. (Output placement admits the dual form: g pinned to one on retained coordinates.) $\square$

A.3 Proof of Theorem 1

Sufficiency. Assume $[P_A, F'](z) = 0$ for all reachable states z of the phase-A trajectory and $P_A x_0 = x_0$. Induction: $h_\ell = F_A^\ell(x_0) \in \text{im } P_A$; then $h_{\ell+1} = P_A F'(h_\ell) = F'(P_A h_\ell) = F'(h_\ell)$, whose relevant components equal $F_A(h_\ell)$ since the absent suffix contribution cannot affect them under the commutator condition; hence $h_{\ell+1} = F_A^{\ell+1}(x_0)$. *Necessity (hard-projection class).* If $[P_A, F'](z^*) \neq 0$ at some reachable z^* , the first passage through z^* yields $P_A F'(z^*) \neq F'(z^*)$: a needed component drops or a foreign one survives, and the deviation persists (and grows by Proposition 2). No depth-constant projection repairs it post hoc. $\square$

A.4 Proof sketch of Proposition 1

Two-dimensional counterexample with $I_A = \{1\}$, $I_B = \{2\}$, $\sigma = \text{ReLU}$: $f'_1(x) = \sigma(x_1) + \varepsilon x_2$, $f'_2(x) = \delta \sigma(x_1) + \gamma x_2$, $\varepsilon \delta \neq 0$. Preservation from $x_0 = (a, 0)$ requires $(1+g_1)^2 + g_1 g_2 \varepsilon \delta = 4$ for all $a > 0$; with $g_2 > 0$ and σ non-affine this has no constant solution (g_1, g_2) independent of a , and any a -dependent relaxation already deviates in the intermediate state. For affine f' the equation is solvable—nonlinearity is exactly what forces hardness. $\square$

A.5 Proof of Proposition 2

Error recursion $e_{\ell+1} = e_\ell + [f'(\tilde{h}_\ell) - f'(h_\ell)] + \delta_\ell$ with $\|f'(\tilde{h}) - f'(h)\| \leq L_f \|e_\ell\|$ gives $\|e_L\| \leq \sum_j (1 + L_f)^{L-1-j} \delta_j$. Uniform bounds require $\rho < 1$ on the interference dynamics; $\rho > 1$ compounds geometrically. Empirical directional gains 1.05–1.89 place trained BDH in the expansive regime. $\square$

A.6 Remark (LayerNorm)

Commutation holds iff suffix pre-LN contributions vanish exactly; LayerNorm mixes statistics globally, so soft activity breaks it at first order. The measured leakage curve (main text, Fig. 3) is the empirical norm of this broken commutator, and the $j \leq 0.15$ budget is its operating consequence.

B Prior-art triage

To our knowledge no prior work states or resolves forward-path isolation for a frozen operator reused across depth via additive growth plus selection.

C Reproducibility

Model: BDH, 100,925,440 parameters ($d=512$, $n_h=8$, $L=6$, multiplier 128, block 512, byte vocabulary); bf16 autocast; AdamW peak lr 10^{-3} cosine-decayed; warmup 1000; batch 4×512 tokens. Corpora: Europarl v7 (30 MB caps; FR/PT pairs downloaded and integrity-checked) and a three-register textmix (Wikipedia/Gutenberg/parliamentary, 28 MB train each after holdout carving). Seeds $\{1337, 1, 2\}$ cover the core pipeline; all other runs single-seed as noted. Every table’s numbers trace to logged runs under version control; the companion report (`docs/reports/2026-08-25_cl-h1-report.md`, §1–20) carries complete intermediate tables, and all mechanisms ship in the repository (`scripts/`: likelihood router, merge/prune sweeps, importance extraction, gated distillation).

line of work	what it isolates	same problem?
Progressive nets / PathNet	parameter paths (removes sharing)	no – bypasses sharing
PackNet / supermasks	pruned-and-frozen weight subsets	no – weight space only
HAT	task masks over weights/gradients	close, but not forward-path
EWC / SI / MAS	quadratic weight penalties	no – importance does not localize here
OGD / GPM / A-GEM	gradient-subspace projection	no – optimizer geometry
Hypernetworks	generated per-task weights	no – different mechanism
Adapters / LoRA	low-rank add-ons	partially; no depth-recurrence analysis
MoE / conditional computation	learned expert mixing	vocabulary matches; experts jointly trained
Highway / GateSkip gating	scaled residual updates	existence evidence; per-layer params
ResCL / branch mixing	old/new branches retrained	violates frozen path
Modular nets / DIB	routed modules	empirical results, no theorem
Upstream BDH fork (PR)	EWC + gates, Permuted-MNIST scale	protection class on disjoint-regime toys